

1 FindBack Technical Debt Plan

FindBack was deliberately scoped for a 48-hour individual capstone. The following debt is visible, classified, and planned rather than concealed. No critical debt is knowingly left unresolved in the implemented privacy, authentication, role-check, or allowed-status rules.

Debt	Cause	Impact	Priority	Classification	Proposed resolution and release
In-app-only claim alerts	The time-box excluded e-mail, push, and SMS integrations.	A claimant must return to FindBack to read a decision.	Medium	Scheduled	Add verified e-mail/push channels and notification preferences in v1.1.
One image per report	A single upload path reduces storage and validation complexity.	A student cannot add multiple angles or supporting images.	Low	Temporarily acceptable	Add an <code>item_images</code> relation, gallery UI, and file-ordering support in v1.1.
Basic relational foreign-key enforcement	The initial schema prioritises rapid, portable deployment.	Orphaned references would require application-level safeguards if data is changed outside the application.	Medium	Scheduled	Add FK constraints and deletion/retention rules after confirming migration compatibility in staging.
Simplified keyword search	The expected examination data volume is small.	No ranking, typo tolerance, synonym matching, or full-text index.	Low	Temporarily acceptable	Add database full-text search, ranking, and searchable field indexes in v1.2.

Debt	Cause	Impact	Priority	Classification	Proposed resolution and release
Manual content moderation	The project does not include a paid or trained moderation service.	Administrators must recognise unsuitable text/images.	Medium	Scheduled	Add report-abuse workflow, moderation queue, audit log, and content-policy review before wider rollout.
Manual acceptance-test evidence	Separate accounts and a deployed URL are needed for authentic end-to-end validation.	The final manual test table is incomplete until deployment.	Medium	Requires attention before submission	Execute the recorded live checklist with one student and one admin account, then attach final evidence.
Single deployable bundle	A 48-hour project uses a compact frontend bundle.	Initial load is larger than an enterprise-scale app should target.	Low	Scheduled	Add route-level dynamic imports and assess bundle-size improvements in v1.1.

1.1 Repayment Order

The project must first complete the manual live test evidence because it affects examination readiness. The next release should introduce notification delivery and data-integrity constraints because they improve operational reliability. Search, multi-image support, moderation, and code splitting follow in value-driven increments, informed by real usage and administrator feedback.

1.2 Maintenance Link

Technical-debt reviews will occur during each release-planning cycle. Every debt item will be reassessed for impact, priority, owner, due release, and verification result. New shortcuts must be added to this register before release approval.